INCH-POUND
MIL-PRF-43709C
2 March 2017
SUPERSEDING
MIL-T-43709B
13 November 1990

DETAIL SPECIFICATION

TAPE, TEXTILE, ARAMID

This specification is approved for use by all Departments and Agencies of the Agencies of the Department of Defense (DoD).

1. SCOPE

1.1 Scope. This specification covers one type of meta-aramid nylon tape.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in section 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in section 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed: ATTN: DLA Troop Support, 700 Robbins Avenue, Philadelphia, PA 19111-5096. Since contact information can change, you may want to verify the currency of the address information using Acquisition Streamlining and Standardization Information System (ASSIST) online database https://assist.dla.mil.

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2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified. The issues are those cited in the solicitation or contract.

FEDERAL TRADE COMMISSION

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies are available online at http://www.ftc.goc.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC Test Method 16.2 - Colorfastness to light: Carbon Arc
AATCC Test Method 16.3 - Colorfastness to Light: Xenon
AATCC Test Method 20 - Fiber Analysis: Qualitative
AATCC Test Method 61 - Colorfastness to Laundering: Accelerated
AATCC Test Method 81 - pH of Water-Extract from Wet Processed Textiles
AATCC Evaluation Procedure 1, Gray Scale for Color Change
AATCC Evaluation Procedure 2, Gray Scale for Staining
AATCC Evaluation Procedure 9, Visual Assessment of Color Difference of Textiles

(Copies are available online at http://www.aatcc.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ANSI/ASQ Z1.4 – Sampling Procedures and Tables for Inspection by Attributes

(Copies are available online at http://www.asq.org.)

ASTM INTERNATIONAL

ASTM D276 - Standard Test Method for Identification of Fibers in Textiles
ASTM D1907/D1907M - Standard Test Method for Linear Density of Yarn (Yarn Number) by the Skein Method
ASTM D3775 - Standard Test Method for Fabric Count of Woven Fabric
ASTM D3776/D3776M - Standard Test Method for Mass Per Unit Area (Weight) of fabric
ASTM D5035 - Standard Test Method for Breaking Force and Elongation of Textile Fabrics (Strip Method)

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(Copies of documents are available online at http://www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Standard Sample. The finished tape shall match the standard sample for shade and appearance, and shall, unless otherwise indicated, be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.3 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Materials.

3.4.1 Yarn. The yarn for the warp and filling shall be made from a bright continuous filament meta-aramid fiber with a nominal denier of 200 for the warp and filling.

3.4.2 Yarn (alternate construction). The yarn for the warp and filling shall be made from meta-aramid spun into 40/2 (nominal) cotton count.

3.5 Color. The color of the tape shall be a good match to the standard sample for Tan 499 or as specified in the contract or purchase order (see 6.2) when tested in accordance with 4.6.

3.5.1 Colorfastness. The finished tape shall conform to the colorfastness requirements listed in Table I, unless otherwise stated in the contract or procurement documents when tested as specified in 4.6.

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TABLE I. Colorfastness requirements.

Color Evaluation	Light (after 40 AFU or 170 kJ/(m^2nm)@420nm), (min.) 1/	Laundering (4 cycles), (min.)
Tan 499	3-4	3-4
All colors	3-4	3-4

1/ AFU: AATCC Fading Units

3.6 Physical requirements. The finished tape shall conform to the requirements, listed in Table II, when tested as specified in 4.6.

TABLE II. Physical requirements.

	Requirement	
Characteristic	Multifilament Yarn	Spun Yarn (alternate)
Width (inch)	1/4 (±1/64)	1/4 (±1/64)
Weight, oz./lin.yd. (max.)	0.07	0.10
Breaking strength, lb. (min.)	100	75
Yarns (min.) Warp, full width Filling yarns per inch	 58 32	 46 36

3.7 Weave. The weave for the tape shall be a plain weave, except that the warp ends shall weave two ends as one. The filling yarn shall traverse the full width of the tape with one (1) filling yarn per shed.

3.8 pH. The pH value of the water extract of the finished tape shall not be less than 5.5 or more than 9.0 when tested as specified in Table IV.

3.9 Length and put-up. For Government procurement only, unless otherwise specified (see 6.2), the tape shall be furnished in lengths of 500 (± 10)-yards. No spool or tube shall contain more than five (5) pieces, and no piece shall be less than 5-yards in length.

3.10 Fiber identification. Each spool or tube of tape shall be labeled, ticketed, or invoiced for fiber content in accordance with the Rules and Regulations under the Textile Fiber Products Identification Act.

3.11 Workmanship. The finished tape shall conform to the quality of product established by this specification. The quality acceptance level shall not exceed the established maximum value.

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2)
b. Conformance inspection (see 4.3)

4.2 First article inspection. A first article submitted in accordance with 3.1, shall be inspected, examined for appearance, color and finished defects and tested for the characteristics as specified in 4.5 through 4.7.

4.3 Conformance inspection. Conformance inspection shall include the examination of 4.5 and the tests of 4.6 through 4.7 as applicable. Sampling for inspection shall be performed in accordance with ANSI/ASQ Z1.4 and with quality acceptance limits as specified in the contract and/or order, except where otherwise indicated (see 6.2).

4.4 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with all the requirements of referenced documents, unless otherwise excluded, amended, modified or qualified in this specification or applicable procurement documents (see 6.2).

4.5 Visual examination. The tape shall be examined for the defects listed in Table III.

TABLE III. Visual defects.

		Classification	
Examine	Defect	Major	Minor
Weave	Uneven weaving throughout		201
Twist or distortion	Webbing or tape will not lay flat upon application of manual pressure due to twist or distortion. 1/		202
Cut, hole, or tear	Any cut, hole, or tear.	101	
Spot, stain, or streak	Any clearly visible 2/		203
Broken or missing end or pick	End: Two or more regardless of length or a single exceeding 6-inches in length.	102	
	Pick: Two or more regardless of length.		204
Edges	Frayed, slack or otherwise poorly constructed and exceeding 1/4-inch in length.	103	
Mispick, double pick	Two or more across the full width.	105	
Width	Beyond specified tolerances		205

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TABLE III. Visual defects. - Continued

		Classification	
Examine	Defect	Major	Minor
Color	Wrong color.		206
Label	Mislabeled, not labeled in accordance with the Rules and Regulations under the Textile Fiber Products Identification Act.	106	
Length	Gross length of spool or tube not as specified; is more or less than 2-yards marked on the spool or tube	107	
	Individual tape piece is less than 5-yards or more than five (5) pieces on a spool or tube.	108	

1/ A 3-yard length of tape/webbing shall be laid on a flat and smooth surface without tension. If the tape does not lie flat, or if the tape is still wavy or ridgy after applying manual pressure, it shall be counted as a defect.
2/ Clearly visible at normal inspection distance of three (3)-feet.

4.6 End item testing. The tape shall be tested for the characteristics listed in Table IV. The methods of testing as specified wherever applicable and as listed in Table IV shall be followed. All test reports shall contain the individual values utilized in expressing the final results. The sample unit for testing shall be ten (10) continuous yards of the finished tape for all physical and chemical tests. The lot shall be unacceptable if one or more sample units fail to meet any requirement specified. The sample size shall be in accordance with the following:

Lot size (yards)	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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TABLE IV. End item tests.

Characteristics	Requirement Reference	Test Method
Yarn identification	3.4.1, 3.4.2	AATCC 20 or ASTM D276, 1/, (see 6.4)
Denier	3.4.1	ASTM D1907/D1907M
Yarn cotton count	3.4.2	Visual, 1/, (see 6.4)
Visual shade matching	3.5	4.7.1
Colorfastness to: Light Laundering	 Table I Table I	 AATCC 16.2, Option 1 or 16,3, Option 2, 2/ AATCC 61, Test 3A, 3/
Weight	Table II	ASTM D3776/D3776M, Option D
Breaking strength	Table II	ASTM D5035
Yarns Warp, full width Filling yarns per inch	 Table II Table II	 ASTM D3775 ASTM D3775
Weave	3.7	Visual
pH	3.8	AATCC 81

1/ In case of dispute, the ASTM method prevails.
2/ Rated using the AATCC Evaluation Procedure 1, Gray Scale for Color Change.
3/ Rated using the AATCC Evaluation Procedure 1, Gray scale for Color Change and AATCC Evaluation Procedure 2, Gray scale for Staining.

4.7 Methods of inspection.

4.7.1 Visual shade matching. The color and appearance of the webbing and tape shall match the standard sample when viewed using the ATTCC Evaluation Procedure 9, Option A, with sources simulating artificial daylight D75 illuminant with a color temperature of 7500K (± 200) illumination of 100 (± 20) foot candles, and shall be a good match to the standard sample under incandescent A illuminant with a color temperature of 2856K (± 200).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or purchase order (see 6.2). When actual packaging of material is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Department or Defense Agency, or within the military service's system commands. Packaging data retrieval

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is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory or contractually binding).

6.1 Intended use. The tape is intended for use as a slide fastener pull thong for Combat Vehicle Crewmen (CVC) uniforms.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this Specification
b. The specific issue of individual documents referenced (see 2.2)
c. When a first article is required (see 3.1 and 4.2)
d. Color required (see 3.5)
e. Length required if other than specified (see 3.9)
f. Conformance inspection acceptance quality limits (AQL) (see 4.3)
g. Inspection conditions (see 4.4)
h. Packaging (see 5.1)

6.3 Standard sample. For access to samples and pattern drawings, address the contracting activity issuing the invitation for bids or request for proposal.

6.4 Certificate of compliance. The contracting activity may select to accept a certificate of compliance for stated requirement.

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issues, due to the extensiveness of the changes.

6.6 Subject term (key word) listing.

Combat Vehicle Crewmen
Meta-aramid
Uniforms
Slide fastener pull
Thong
Yarn
Weave

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Custodian:
Army – GL

Preparing Activity
DLA-CT

Review Activities:
Army – MD
Navy – NU

(Project 8315-2017-002)

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